

E-Commerce Customer Segmentation — Who Are Our Customers?

Industry: Retail and E-Commerce **Algorithm:** KNN

Note: The dataset is synthetic and was created for analytical demonstration.

1. Problem Statement

The company wants behaviour-based customer segmentation using purchase frequency, recency, spending, product diversity and discount usage.

2. Objectives

Clean data, calculate RFM metrics, identify meaningful segments, compare customer groups, identify frequent low-spending and heavy infrequent customers, build KNN, evaluate it and translate segments into marketing actions.

3. Data Cleaning

Missing numeric values were imputed with medians, duplicates were checked and removed, and descriptive statistics were used to validate the dataset.

4. RFM Method

Recency was scored from 5 (most recent) to 1 (least recent). Frequency and spending were scored from 1 to 5. The scores were combined into an RFM score.

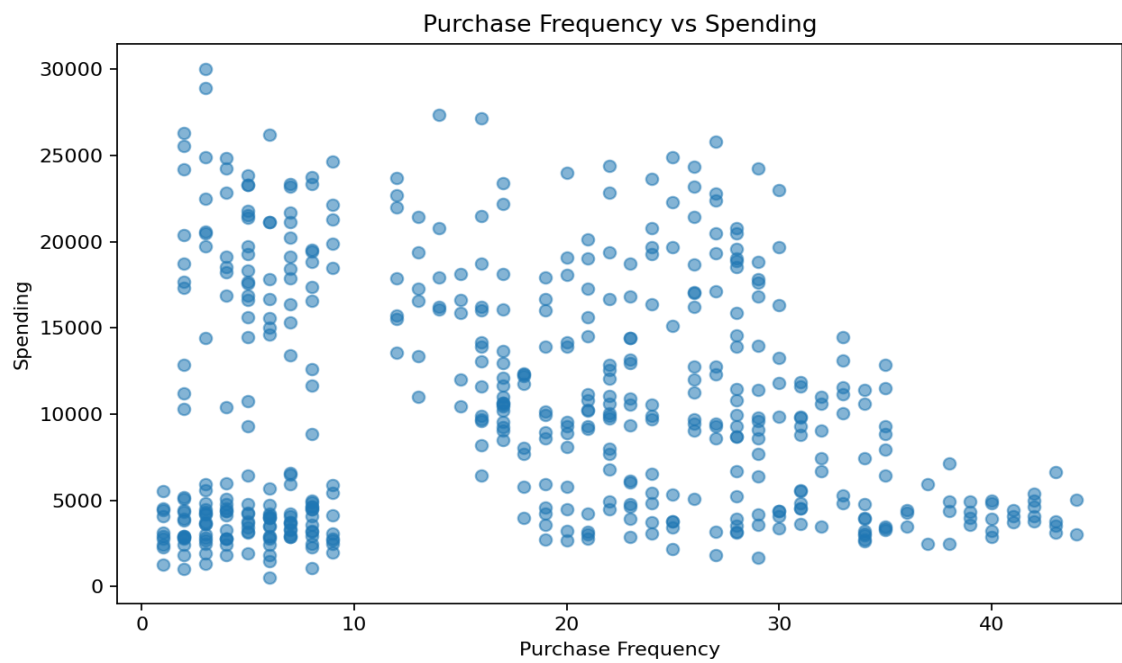
5. Segment Summary

Segment	Customers	Avg Recency	Avg Frequency	Avg Spending
Heavy Infrequent	80	86.21	6.05	19583.83
High Value	46	12.46	26.78	17712.34
Loyal / Potential	188	26.00	21.38	11438.64
Frequent Low Spend	79	17.84	33.04	3950.99
At Risk / Low Value	107	115.64	4.83	3447.29

6. Visualization 1



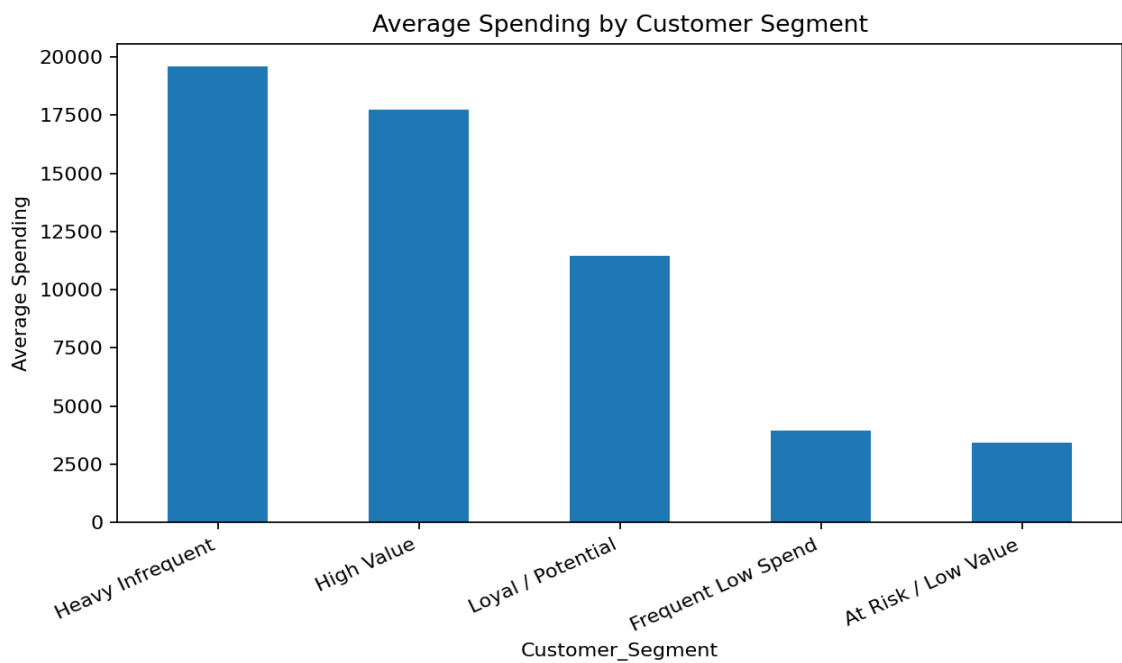
7. Visualization 2



8. Visualization 3



9. Visualization 4



10. KNN Model and Evaluation

A KNN classifier with $k=7$ was trained on standardized Recency, Purchase Frequency, Spending, Product Diversity and Discount Usage. The target was the behaviour-based customer segment. The data was split 80/20 with stratification.

Metric	Score
Accuracy	0.920
Precision	0.919
Recall	0.920
F1 Score	0.916

11. Behaviour Pattern Flagging

Average distances to the five nearest neighbours were calculated after standardization. The highest 5% were flagged as unusual behaviour patterns, resulting in 25 flagged customers.

12. Key Insights

1. RFM analysis separates customers into distinct behaviour-based groups.
2. High-value customers combine recency, frequency and spending strength.
3. Frequent low-spending customers are opportunities for basket-size growth.
4. Heavy-spending but infrequent customers are valuable reactivation targets.
5. At-risk/low-value customers require selective win-back activity.
6. Product diversity and discount usage provide additional targeting context.
7. KNN supports similarity-based classification and nearest-neighbour distance can flag unusual behaviour.

13. Practical Marketing Action Plan

Segment	Action
High Value	VIP rewards, early access and personalized offers.
Loyal / Potential	Cross-sell, bundles and loyalty benefits.
Frequent Low Spend	Bundles and threshold-based offers to increase basket size.
Heavy Infrequent	Personalized reactivation and recommendations.
At Risk / Low Value	Low-cost win-back campaigns and selective incentives.

14. Conclusion

RFM-based segmentation provides a behaviour-focused customer view beyond broad demographics. KNN adds similarity-based classification, while nearest-neighbour distance helps flag unusual behaviour patterns. These outputs can support targeted marketing actions.